ClipSAT : Egyptian Scholastic Test: EST II Mathematics Level 2 [ANSWER KEY]

Unit 2.01: Sequences, Series & Sigma Notation

Q1. Consider an arithmetic sequence where the first term is a_1 = 5 and the common difference is d = 3 . What is the value of the 4th term, a_4 ?

[x] A: 14
[ ] B: 17
[ ] C: 8
[ ] D: 11

Solution: The formula for the nth term of an arithmetic sequence is given by a_n = a_1 + n-1d . In this problem, we are given a_1 = 5 and d = 3 . We need to find the 4th term, so n = 4 . Plugging these values into the formula, we get: a_4 = 5 + 4-1 times 3 a_4 = 5 + 3 times 3 a_4 = 5 + 9 a_4 = 14 . Therefore, the 4th term of the sequence is 14.

Q2. A geometric sequence has a_1=6 and common ratio r=2 . Find a_{5} .

[x] A: 96
[ ] B: 192
[ ] C: 98
[ ] D: -96

Solution: Applying Sequences & Series principles, a_{5}=a_1r^{n-1}=62^{4}=96 . Therefore, the verified correct option is 96.

Q3. Consider a geometric sequence with positive common ratio r > 0 where the first term is a_1 = frac{1}{2} . If the sum of the first three terms is S_3 = frac{7}{2} , find the value of r .

[x] A: 2
[ ] B: -2
[ ] C: 1/2
[ ] D: 3

Solution: The sum of the first three terms is a_11 + r + r^2 = frac{7}{2} . Substituting a_1 = frac{1}{2} gives: 1 + r + r^2 = 7 implies r^2 + r - 6 = r + 3r - 2 = 0. Since r > 0 , the common ratio is r = 2 .

Q4. A geometric sequence has a_1=2 and common ratio r=3 . Find a_{5} .

[x] A: 162
[ ] B: 486
[ ] C: 165
[ ] D: -162

Solution: Applying Sequences & Series principles, a_{5}=a_1r^{n-1}=23^{4}=162 . Therefore, the verified correct option is 162.

Q5. A geometric sequence has a_1=6 and common ratio r=3 . Find a_{5} .

[x] A: 486
[ ] B: 1458
[ ] C: 489
[ ] D: -486

Solution: Applying Sequences & Series principles, a_{5}=a_1r^{n-1}=63^{4}=486 . Therefore, the verified correct option is 486.

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Q6. A geometric sequence has a_1=2 and common ratio r=-2 . Find a_{4} .

[x] A: -16
[ ] B: 32
[ ] C: -18
[ ] D: 16

Solution: a_{4}=a_1r^{n-1}=2-2^{3}=-16 .

Q7. A geometric sequence has a_1=5 and common ratio r=3 . Find a_{3} .

[x] A: 45
[ ] B: 135
[ ] C: 48
[ ] D: -45

Solution: Applying Sequences & Series principles, a_{3}=a_1r^{n-1}=53^{2}=45 .
Therefore, the verified correct option is 45.

Q8. A geometric sequence has a_1=1 and common ratio r=-3 . Find a_{5} .

[x] A: 81
[ ] B: -243
[ ] C: 78
[ ] D: -81

Solution: Applying Sequences & Series principles, a_{5}=a_1r^{n-1}=1-3^{4}=81 .
Therefore, the verified correct option is 81.